

**A Field Project Report on
E-Commerce Website**

Submitted

In partial fulfillment of the requirements for the award of the degree

**BACHELOR OF TECHNOLOGY
in**

**COMPUTER SCIENCE and ENGINEERING
by**

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VIGNAN'S

FOUNDATION FOR SCIENCE, TECHNOLOGY & RESEARCH

(Deemed to be University) - Estd. u/s 3 of UGC Act 1956

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

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CERTIFICATE

This is to certify that the field project entitled "E-Commerce Website" is being submitted by [CH Bala Sai] , [231FA04E77] , [P Sai Nuthan] , [231FA04F21], [V Janaki] , [231FA04F52] , [C Lokesh] , [231FA04G70] in partial fulfilment of the requirements for the degree of Bachelor of Technology (B.Tech.) in Computer Science and Engineering at the Department of Computer Science and Engineering, Vignan's Foundation for Science, Technology and Research (Deemed to be University), Vadlamudi, Guntur District, Andhra Pradesh, India.

This is a bonafide work carried out by the aforementioned students under my guidance and supervision.


Guide



Project Review Committee


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DECLARATION

Date: 26/04/2025

We hereby declare that the work presented in the field project titled "E-Commerce Website" is the result of our own efforts and investigations.

This project is being submitted under the supervision of **Ch. Swarna Lalitha Asst.prof,** Department of CSE in partial fulfillment of the requirements for the Bachelor of Technology (B.Tech.) degree in Computer Science and Engineering at Vignan's Foundation for Science, Technology and Research (Deemed to be University), Vadlamudi, Guntur, Andhra Pradesh, India

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1.Introduction

An e-commerce website is an online platform that facilitates the buying and selling of goods and services over the internet, enabling businesses and individuals to conduct transactions through digital storefronts accessible via computers, tablets, smartphones, and other smart devices. These websites typically feature product catalogs, search functionality, user reviews, secure payment gateways, and order tracking, providing a convenient and efficient shopping experience for users.

1.1 Problem Definition

An e-commerce website aims to address the challenges faced by businesses and consumers in the online retail space by providing a comprehensive, user-friendly, and secure platform for buying and selling products and services. Such a platform should facilitate seamless transactions, offer personalized shopping experiences, and ensure efficient management of product information and customer data.

1.2 Existing Systems

1.2.1 Standalone E-commerce Platforms

- **Shopify:** Shopify offers a comprehensive e-commerce solution with integrated AI tools, enabling merchants to manage their online stores effectively.
- **Magento:** Magento provides a flexible e-commerce platform with extensive customization options, catering to businesses of various sizes.
- **WooCommerce:** WooCommerce is a WordPress plugin that transforms websites into fully functional e-commerce stores, offering a range of plugins and themes.

1.2.2 Integrated E-commerce Solutions

- **Amazon:** Amazon's platform allows third-party sellers to list products alongside Amazon's own inventory, providing a vast marketplace for consumers.
- **eBay:** eBay facilitates auctions and direct sales, connecting buyers and sellers globally across various product categories.

1.3 Proposed System

The proposed e-commerce website aims to enhance the online shopping experience by integrating advanced search functionalities, personalized recommendations, secure payment systems, and efficient order management. By incorporating features such as responsive design, mobile optimization, and AI-driven analytics, the platform seeks to meet the evolving needs of modern consumers.

1.4 Literature Review

Existing e-commerce platforms offer a range of features, but there are opportunities to improve in areas such as user experience, personalization, and system scalability. Studies suggest that integrating AI technologies can significantly enhance product recommendations and customer engagement. Additionally, adopting modular system designs can improve scalability and flexibility, allowing businesses to adapt to changing market demands.

2. System Requirements

2.1 Hardware and Software Requirements

- **Hardware:** Servers with high availability, load balancing, and sufficient storage capacity to handle peak traffic and transaction volumes.
- **Software:** E-commerce platform software (e.g., Shopify, Magento), database management systems (e.g., MySQL, PostgreSQL), programming languages (e.g., PHP, Python), and cloud services for scalability.

2.2 Software Requirements Specifications (SRS)

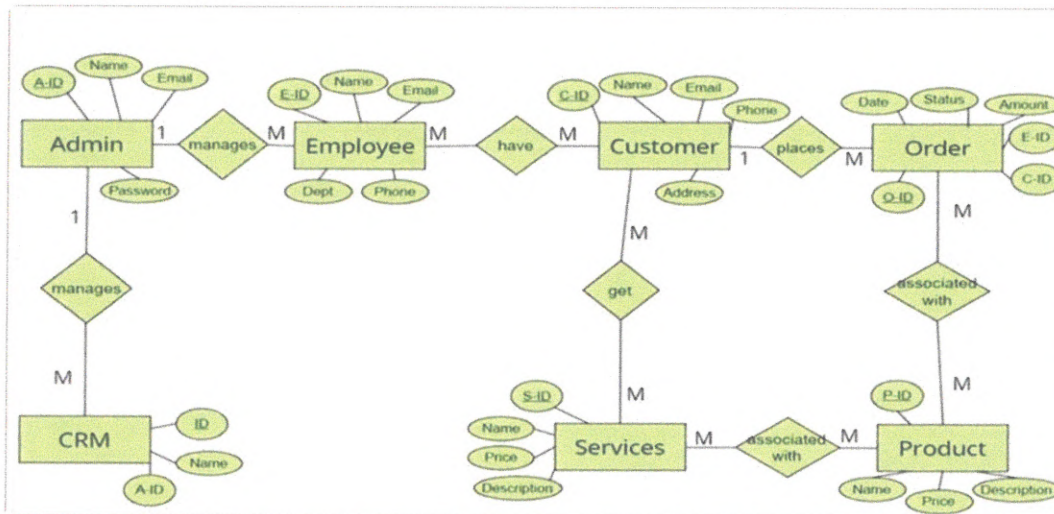
The SRS document will outline the functional and non-functional requirements of the e-commerce system, including user authentication, product catalog management, shopping cart functionality, payment processing, and order tracking. It will also define performance metrics, security protocols, and compliance with data protection regulations.

3. System Design

3.1 Model of the System

Implementing a modular architecture in an e-commerce website enhances scalability, flexibility, and maintainability by dividing the platform into independent modules. This approach allows for focused development, seamless integration of new features, and efficient updates without disrupting the entire system. Such a design is crucial for adapting to evolving business needs and delivering a seamless user experience.

3.2 E.R Diagram



4. Implementation

4.1 Sample code

Index.Html

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Premium Kitchen Solutions - Glen, Hafele, Crompton</title>

  <link rel="stylesheet" href="styles.css" />

</head>

<body>

<header>

  <h1>Premium Kitchen Solutions</h1>

  <p>Glen • Hafele • Crompton</p>

</header>

<nav>

  <a class="profile-link" onclick="showProfile()">My Account</a>

  <div class="nav-links">

    <a href="#" onclick="showSection('home')">Home</a>

    <a href="#" onclick="showSection('auth')">Login / Register</a>

    <a href="#" onclick="showSection('products')">Products</a>

    <a href="#" onclick="showSection('cart')">Cart</a>

    <a href="#" onclick="showSection('sell')">Sell With Us</a>
```

```

    <a href="#" onclick="showSection('about')">About Us</a>
    <a href="#" onclick="showSection('help')">Help</a>
</div>
</nav>

<div class="container">

    <!-- Home Section -->
    <div class="section" id="home">
        <div class="search-bar">
            <input type="text" id="search-input" placeholder="Search stoves, chimneys...">
            <button onclick="searchProduct()">Search</button>
        </div>
        <h2>Premium Kitchen Appliances</h2>
        <p>Discover the finest stoves and chimneys from Glen, Hafele, and Crompton for your modern kitchen.</p>

        <!-- Search Results Section -->
        <div class="search-results" id="search-results">
            <!-- Search results will be displayed here -->
        </div>

        <!-- Display Products on Home Page -->
        <h3>Featured Products</h3>
        <div class="product-list" id="home-product-list">
            <!-- Products will be dynamically added here -->
        </div>
    </div>

    <!-- Login / Sign In Section -->

```

```

<div class="section" id="auth">
  <form class="auth-form" id="auth-form" onsubmit="handleAuth(event)">
    <h2 id="auth-title">Login</h2>
    <input type="email" id="auth-email" placeholder="Enter your email" required>
      <input type="password" id="auth-password" placeholder="Enter your password"
required>
    <button type="submit" id="auth-button">Login</button>
    <div class="toggle-link" onclick="toggleAuth()">Don't have an account? Register</div>
  </form>
</div>

```

```

<!-- Product Section -->
<div class="section" id="products">
  <h2>Our Products</h2>
  <div class="brand-filter">
    <button onclick="filterProducts('all')" class="active">All Brands</button>
    <button onclick="filterProducts('glen')">Glen</button>
    <button onclick="filterProducts('hafele')">Hafele</button>
    <button onclick="filterProducts('crompton')">Crompton</button>
  </div>
  <div class="product-list" id="product-list">
    <!-- Products will be dynamically added here -->
  </div>
</div>

```

```

<!-- Sell Product Section -->
<div class="section" id="sell">
  <form class="sell-form" id="sell-form" onsubmit="handleSell(event)">
    <h2>Sell With Us</h2>
    <label for="sell-name">Product Name</label>

```

<input type="text" id="sell-name" placeholder="e.g., Glen 4 Burner Gas Stove" required>

<label for="sell-brand">Brand</label>

<select id="sell-brand" required>

<option value="">Select Brand</option>

<option value="Glen">Glen</option>

<option value="Hafele">Hafele</option>

<option value="Crompton">Crompton</option>

</select>

<label for="sell-type">Product Type</label>

<select id="sell-type" required>

<option value="">Select Type</option>

<option value="Gas Stove">Gas Stove</option>

<option value="Induction Cooktop">Induction Cooktop</option>

<option value="Kitchen Chimney">Kitchen Chimney</option>

<option value="Hob">Hob</option>

</select>

<label for="sell-price">Price (₹)</label>

<input type="number" id="sell-price" placeholder="Enter price in INR" required>

<label for="sell-image">Product Image URL</label>

<input type="url" id="sell-image" placeholder="https://example.com/image.jpg" required>

<label for="sell-desc">Product Description</label>

<textarea id="sell-desc" placeholder="Enter product features and description"></textarea>

```
<button type="submit">Submit Product</button>
</form>
</div>
```

```
<!-- Cart Section -->
<div class="section" id="cart">
  <h2>Your Shopping Cart</h2>
  <div class="cart-items" id="cart-items">
    <!-- Cart items will be displayed here -->
  </div>
  <div class="cart-total" id="cart-total">Total: ₹0</div>
  <button onclick="checkout()">Proceed to Checkout</button>
```

```
<!-- Checkout Log Section -->
<h3>Checkout History</h3>
<textarea id="checkout-log" readonly style="width:100%; height:150px; font-size:0.9rem;
padding:10px; margin-top:1rem; border-radius:6px; border:1px solid #ccc;
background:#fdfdfd;"></textarea>
</div>
```

```
<!-- About Us Section -->
<div class="section" id="about">
  <h2>About Premium Kitchen Solutions</h2>
  <div class="about-content">
    <h3>Our Story</h3>
    <p>Premium Kitchen Solutions is India's leading online destination for premium kitchen
appliances. We specialize in offering the finest products from trusted brands like Glen,
Hafele, and Crompton to transform your cooking experience.</p>
    <h3>Our Brands</h3>
```

Glen: Known for its innovative and stylish kitchen appliances, Glen offers a wide range of gas stoves, induction cooktops, and chimneys that combine performance with elegance.

Hafele: A global leader in kitchen solutions, Hafele brings German engineering to your kitchen with their premium hobs, chimneys, and modular kitchen accessories.

Crompton: With decades of trust, Crompton offers reliable and energy-efficient kitchen appliances that cater to every Indian household's needs.

Our Mission

We aim to provide our customers with the best kitchen solutions that enhance their cooking experience while offering excellent value for money. Our team carefully selects each product to ensure quality, durability, and performance.

<!-- Help Section -->

<div class="section" id="help">

<h2>Help & Support</h2>

<div class="help-content">

<h3>Frequently Asked Questions</h3>

Q: How do I place an order?

A: Simply browse our products, add items to your cart, and proceed to checkout.

Q: What payment methods do you accept?

A: We accept all major credit/debit cards, net banking, UPI, and cash on delivery.

Q: What is your return policy?

A: We offer a 7-day return policy for unused products in original packaging. Please contact our support team for return requests.

Q: Do you provide installation services?

A: Yes, we offer professional installation services for chimneys and hobs. This can be selected during checkout.</p>

```
<div class="contact-info">
  <h3>Contact Us</h3>
  <div><strong>Phone:</strong> +91 9876543210</div>
  <div><strong>Email:</strong> support@premiumkitchensolutions.com</div>
  <div><strong>Address:</strong> 123 Kitchen Street, Appliance City, Mumbai -
400001</div>
  <div><strong>Working Hours:</strong> Monday-Saturday, 9:00 AM to 6:00 PM</div>
</div>
</div>
</div>
```

```
<!-- Profile Section -->
<div class="profile-section" id="profile-section">
  <h3>My Account <span class="close-profile" onclick="hideProfile()">✕</span></h3>
  <p><span>Email:</span> <span id="profile-email">guest@example.com</span></p>
  <p><span>Account    Type:</span>    <span    id="account-type">Regular
Customer</span></p>
  <p><span>Orders:</span> <span id="order-count">0</span> placed</p>
  <button onclick="logout()">Logout</button>
</div>
```

```
</div>
```

```
<script src="script.js"></script>
```

```
</body>
```

```
</html>
```

Styles.css

```
/* General Styles */
```

```
* {  
  margin: 0;  
  padding: 0;  
  box-sizing: border-box;  
}
```

```
body {  
  font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;  
  background-color: #f8f8f8;  
  color: #333;  
}
```

```
header {  
  background-color: #2c3e50;  
  color: white;  
  padding: 1.5rem;  
  text-align: center;  
  box-shadow: 0 2px 10px rgba(0,0,0,0.1);  
}
```

```
nav {  
  display: flex;  
  flex-direction: column;  
  background-color: #34495e;  
  padding: 1rem;  
  position: relative;  
}
```

```
.profile-link {  
  align-self: flex-end;
```

```
color: white;
text-decoration: none;
padding: 0.5rem;
cursor: pointer;
background-color: #e74c3c;
border-radius: 4px;
}
```

```
.profile-link:hover {
  background-color: #c0392b;
}
```

```
.nav-links {
  display: flex;
  justify-content: space-between;
  margin-top: 0.8rem;
  flex-wrap: wrap;
}
```

```
nav a {
  color: white;
  text-decoration: none;
  padding: 0.8rem 1rem;
  margin: 0.2rem;
  border-radius: 4px;
  transition: all 0.3s ease;
}
```

```
nav a:hover {
  background-color: #2c3e50;
}
```

```
}
```

```
.container {  
  padding: 2rem;  
  max-width: 1200px;  
  margin: 0 auto;  
}
```

```
.section {  
  display: none;  
}
```

```
.show {  
  display: block;  
}
```

```
.search-bar {  
  display: flex;  
  justify-content: center;  
  margin-bottom: 2rem;  
}
```

```
.search-bar input {  
  padding: 0.8rem;  
  font-size: 1rem;  
  border-radius: 5px 0 0 5px;  
  border: 1px solid #ddd;  
  width: 400px;  
  outline: none;  
}
```

```
.search-bar button {  
  padding: 0.8rem 1.5rem;  
  background-color: #e74c3c;  
  color: white;  
  border: none;  
  border-radius: 0 5px 5px 0;  
  cursor: pointer;  
  font-weight: bold;  
}
```

```
.search-bar button:hover {  
  background-color: #c0392b;  
}
```

```
/* Brand Filter */
```

```
.brand-filter {  
  display: flex;  
  justify-content: center;  
  margin: 1.5rem 0;  
  flex-wrap: wrap;  
}
```

```
.brand-filter button {  
  padding: 0.6rem 1.2rem;  
  margin: 0.5rem;  
  background-color: #ecf0f1;  
  border: none;  
  border-radius: 20px;  
  cursor: pointer;
```

```
font-weight: 600;
transition: all 0.3s ease;
}
```

```
.brand-filter button:hover, .brand-filter button.active {
  background-color: #e74c3c;
  color: white;
}
```

```
/* Search Results */
```

```
.search-results {
  display: grid;
  grid-template-columns: repeat(auto-fill, minmax(280px, 1fr));
  gap: 25px;
  margin-top: 1.5rem;
}
```

```
/* Product and Cart Page */
```

```
.product-list {
  display: grid;
  grid-template-columns: repeat(auto-fill, minmax(280px, 1fr));
  gap: 25px;
  margin-bottom: 2rem;
}
```

```
.product-item {
  background-color: white;
  padding: 20px;
  border-radius: 8px;
  box-shadow: 0 4px 8px rgba(0, 0, 0, 0.1);
}
```

```
text-align: center;
transition: transform 0.3s ease;
position: relative;
}
```

```
.product-item:hover {
  transform: translateY(-5px);
  box-shadow: 0 6px 12px rgba(0, 0, 0, 0.15);
}
```

```
.product-item img {
  width: 100%;
  height: 200px;
  object-fit: contain;
  border-radius: 8px;
  margin-bottom: 1rem;
}
```

```
.product-item h3 {
  font-size: 1.3rem;
  margin: 0.8rem 0;
  color: #2c3e50;
}
```

```
.product-item .brand {
  color: #7f8c8d;
  font-weight: 600;
  margin-bottom: 0.5rem;
}
```

```
.product-item p {  
  font-size: 1.2rem;  
  margin-bottom: 1rem;  
  color: #e74c3c;  
  font-weight: bold;  
}
```

```
.product-item button {  
  background-color: #e74c3c;  
  color: white;  
  padding: 10px 20px;  
  border: none;  
  border-radius: 5px;  
  cursor: pointer;  
  font-size: 1rem;  
  transition: background-color 0.3s ease;  
  width: 100%;  
}
```

```
.product-item button:hover {  
  background-color: #c0392b;  
}
```

```
.product-item .features {  
  text-align: left;  
  margin: 1rem 0;  
  font-size: 0.9rem;  
  color: #7f8c8d;  
}
```

```
.product-item .features li {  
  margin-bottom: 0.3rem;  
  list-style-type: none;  
  position: relative;  
  padding-left: 1.2rem;  
}
```

```
.product-item .features li:before {  
  content: "•";  
  color: #e74c3c;  
  position: absolute;  
  left: 0;  
}
```

```
.cart-items {  
  margin-bottom: 2rem;  
  background-color: white;  
  padding: 1.5rem;  
  border-radius: 8px;  
  box-shadow: 0 4px 8px rgba(0, 0, 0, 0.1);  
}
```

```
.cart-item {  
  display: flex;  
  justify-content: space-between;  
  align-items: center;  
  padding: 1rem 0;  
  border-bottom: 1px solid #eee;  
}
```

```
.cart-item:last-child {  
  border-bottom: none;  
}
```

```
.cart-item span {  
  font-size: 1rem;  
  flex: 1;  
}
```

```
.cart-item .item-name {  
  font-weight: 600;  
}
```

```
.cart-item .item-price {  
  text-align: right;  
  padding-right: 1rem;  
  color: #e74c3c;  
  font-weight: bold;  
}
```

```
.cart-item button {  
  background-color: #e74c3c;  
  color: white;  
  border: none;  
  border-radius: 5px;  
  cursor: pointer;  
  padding: 5px 10px;  
  font-size: 0.9rem;  
}
```

```
.cart-item button:hover {  
  background-color: #c0392b;  
}
```

```
.cart-total {  
  font-size: 1.3rem;  
  font-weight: bold;  
  margin-bottom: 1.5rem;  
  text-align: right;  
  padding: 1rem;  
  background-color: white;  
  border-radius: 8px;  
  box-shadow: 0 4px 8px rgba(0, 0, 0, 0.1);  
}
```

```
.cart button {  
  background-color: #2c3e50;  
  color: white;  
  padding: 12px 25px;  
  border: none;  
  border-radius: 5px;  
  cursor: pointer;  
  font-size: 1.2rem;  
  display: block;  
  margin-left: auto;  
  transition: background-color 0.3s ease;  
}
```

```
.cart button:hover {  
  background-color: #1a252f;
```

```
}
```

```
/* Login/Sign In Form */
```

```
.auth-form {  
  display: flex;  
  flex-direction: column;  
  max-width: 400px;  
  margin: 0 auto;  
  background-color: white;  
  padding: 2rem;  
  border-radius: 8px;  
  box-shadow: 0 4px 8px rgba(0, 0, 0, 0.1);  
}
```

```
.auth-form h2 {  
  text-align: center;  
  margin-bottom: 1.5rem;  
  color: #2c3e50;  
}
```

```
.auth-form input {  
  padding: 1rem;  
  margin-bottom: 1rem;  
  border-radius: 5px;  
  border: 1px solid #ddd;  
  font-size: 1rem;  
}
```

```
.auth-form button {  
  padding: 1rem;
```

```
background-color: #e74c3c;
color: white;
border: none;
border-radius: 5px;
cursor: pointer;
font-size: 1.1rem;
font-weight: 600;
margin-top: 1rem;
}
```

```
.auth-form button:hover {
background-color: #c0392b;
}
```

```
.toggle-link {
text-align: center;
margin-top: 1.5rem;
cursor: pointer;
color: #3498db;
font-weight: 500;
}
```

```
.toggle-link:hover {
text-decoration: underline;
}
```

```
/* Profile Section */
```

```
.profile-section {
display: none;
padding: 1.5rem;
```

```
background-color: white;
border-radius: 8px;
box-shadow: 0 4px 12px rgba(0, 0, 0, 0.15);
position: absolute;
top: 70px;
right: 20px;
width: 300px;
z-index: 100;
}
```

```
.profile-section h3 {
margin-bottom: 1.5rem;
display: flex;
justify-content: space-between;
align-items: center;
color: #2c3e50;
border-bottom: 1px solid #eee;
padding-bottom: 0.8rem;
}
```

```
.close-profile {
cursor: pointer;
color: #7f8c8d;
font-size: 1.5rem;
transition: color 0.3s ease;
}
```

```
.close-profile:hover {
color: #e74c3c;
}
```

```
.profile-section p {  
  margin-bottom: 1rem;  
  color: #34495e;  
}
```

```
.profile-section p span {  
  font-weight: 600;  
  color: #2c3e50;  
}
```

```
.profile-section button {  
  background-color: #e74c3c;  
  color: white;  
  padding: 10px 20px;  
  border: none;  
  border-radius: 5px;  
  cursor: pointer;  
  font-size: 1rem;  
  width: 100%;  
  margin-top: 1rem;  
}
```

```
.profile-section button:hover {  
  background-color: #c0392b;  
}
```

```
/* Sell Product Form */
```

```
.sell-form {  
  display: flex;
```

```
flex-direction: column;
max-width: 500px;
margin: 0 auto;
background-color: white;
padding: 2rem;
border-radius: 8px;
box-shadow: 0 4px 8px rgba(0, 0, 0, 0.1);
}
```

```
.sell-form h2 {
  text-align: center;
  margin-bottom: 1.5rem;
  color: #2c3e50;
}
```

```
.sell-form label {
  margin-bottom: 0.5rem;
  font-weight: 600;
  color: #34495e;
}
```

```
.sell-form input, .sell-form select, .sell-form textarea {
  padding: 1rem;
  margin-bottom: 1.2rem;
  border-radius: 5px;
  border: 1px solid #ddd;
  font-size: 1rem;
}
```

```
.sell-form textarea {
```

```
min-height: 100px;
resize: vertical;
}
```

```
.sell-form button {
padding: 1rem;
background-color: #e74c3c;
color: white;
border: none;
border-radius: 5px;
cursor: pointer;
font-size: 1.1rem;
font-weight: 600;
}
```

```
.sell-form button:hover {
background-color: #c0392b;
}
```

```
/* About Us Section */
```

```
.about-content {
background-color: white;
padding: 2rem;
border-radius: 8px;
box-shadow: 0 4px 8px rgba(0, 0, 0, 0.1);
line-height: 1.6;
}
```

```
.about-content h3 {
color: #2c3e50;
```

```
margin-bottom: 1rem;
}
```

```
.about-content p {
margin-bottom: 1.5rem;
}
```

```
/* Help Section */
```

```
.help-content {
background-color: white;
padding: 2rem;
border-radius: 8px;
box-shadow: 0 4px 8px rgba(0, 0, 0, 0.1);
}
```

```
.help-content h3 {
color: #2c3e50;
margin-bottom: 1rem;
}
```

```
.help-content p {
margin-bottom: 1.5rem;
}
```

```
.contact-info {
margin-top: 2rem;
}
```

```
.contact-info div {
margin-bottom: 1rem;
}
```

```
}
```

```
.contact-info strong {
```

```
  color: #2c3e50;
```

```
}
```

```
/* Responsive adjustments */
```

```
@media (max-width: 768px) {
```

```
  .nav-links {
```

```
    flex-direction: column;
```

```
    align-items: center;
```

```
  }
```

```
  nav a {
```

```
    width: 100%;
```

```
    text-align: center;
```

```
  }
```

```
.search-bar input {
```

```
  width: 70%;
```

```
}
```

```
.product-list, .search-results {
```

```
  grid-template-columns: repeat(auto-fill, minmax(250px, 1fr));
```

```
}
```

```
.profile-section {
```

```
  width: 90%;
```

```
  right: 5%;
```

```
}
```

```
}
```

Script.js

```
const products = [
```

```
{
```

```
  id: 1,
```

```
  name: 'Glen 4 Burner Gas Stove',
```

```
  brand: 'Glen',
```

```
  type: 'Gas Stove',
```

```
  price: 14000,
```

```
    image: 'https://glenindia.com/cdn/shop/files/1_3d518e85-3efc-4d07-82f1-3ddd8a1c4d40_1400x1400.jpg?v=1720501330',
```

```
  features: [
```

```
    '4 brass burners with auto ignition',
```

```
    'Stainless steel body',
```

```
    'Pan support for all vessel sizes',
```

```
    '1-year warranty'
```

```
  ]
```

```
},
```

```
{
```

```
  id: 2,
```

```
  name: 'Hafele 60cm Kitchen Chimney',
```

```
  brand: 'Hafele',
```

```
  type: 'Kitchen Chimney',
```

```
  price: 27000,
```

```
    image: 'https://assets.nikshanonline.com/wp-content/uploads/2024/12/Hafele-60-cm-Wall-Mounted-Chimney-Essentia-Series-RENATA-INCLINED-60-1.png',
```

```
  features: [
```

```
    '60cm width, 1300 m3/hr suction power',
```

```
    'Baffle filter for efficient oil collection',
```

```

    'Touch control panel with 3 speed settings',
    '2-year warranty'
  ],
  {
    id: 3,
    name: 'Crompton 4 Burner Gas Stove',
    brand: 'Crompton',
    type: 'Gas Stove',
    price: 22000,
    image:
      'https://www.crompton.co.in/cdn/shop/files/3d_78_4br.png?v=1727936802&width=1100',
    features: [
      '4 brass burners with auto ignition',
      'Crompton SensoSafe Gas Hob 3D 78cm 4 Burner',
      'Compact design for kitchens',
      '2-year warranty'
    ]
  },
  {
    id: 4,
    name: 'Glen 4 Burner Stainless steel Gas Stove',
    brand: 'Glen',
    type: 'Gas Stove',
    price: 21000,
    image:
      'https://glenindia.com/cdn/shop/files/1_3d518e85-3efc-4d07-82f1-3ddd8a1c4d40_1400x1400.jpg?v=1720501330',
    features: [
      '4 burners with auto ignition',
      'steel top for easy cleaning',

```

```

    'Elegant mat finish',
    '2-year warranty'
  ],
  },
  {
    id: 5,
    name: 'Hafele 4 Burner Hob',
    brand: 'Hafele',
    type: 'Gas stove',
    price: 29000,
    image: 'https://decure.in/cdn/shop/files/hob_haf_verena_470_8098d588-3df7-4834-96c5-e81ffb1334ca.jpg?v=1721805955&width=823',
    features: [
      'Brass Burners',
      'Electric Auto Ignition',
      'Black Glass',
      '2-year warranty'
    ],
  },
  {
    id: 6,
    name: 'Crompton 90cm Kitchen Chimney',
    brand: 'Crompton',
    type: 'Kitchen Chimney',
    price: 18999,
    image: 'https://m.media-amazon.com/images/I/418x49tHKqL._SX679_.jpg',
    features: [
      '90cm width, 1626 m³/hr suction power',
      'Auto clean function',
      'LED lighting',

```

```

    '3-year warranty'
  ]
},
{
  id: 7,
  name: 'Glen Chimney 90cm',
  brand: 'Glen',
  type: 'Kitchen Chimney',
  price: 24000,
  image: 'https://glenindia.com/cdn/shop/products/1_7fc8719f-02bf-4cf3-a978-9b00653a8fc3_1400x1400.webp?v=1673353604',
  features: [
    'Touch controls with motion sensor, wave and start',
    '90cm width, 1400 m³/hr suction power',
    'Auto clean function',
    '1-year warranty'
  ]
},
{
  id: 8,
  name: 'Hafele 90 cm Chimney',
  brand: 'Hafele',
  type: 'Kitchen Chimney',
  price: 23000,
  image: 'https://decure.in/cdn/shop/files/chi_haf_renata_curved_90_dfbd8fed-707e-4fde-a1e0-8a876839e7c5.jpg?v=1721391623&width=823',
  features: [
    '90cm width, 1400 m³/hr suction power',
    'Intelligent Heat Auto Clean with 30 hr timer',
    'European design',

```

'2-year warranty'

]

},

{

id: 9,

name: 'Crompton Hob SensoSafe Plus 80 Cm 4 Burner',

brand: 'Crompton',

type: 'Gas stove',

price: 36000,

image:

'https://www.crompton.co.in/cdn/shop/files/SensoSafeplus80cm4BrPL3.jpg?v=1743171857
&width=1100',

features: [

'Premium mat finish body',

'3D Brass Burners',

'Superior Aesthetics',

'2-year warranty'

]

},

{

id: 10,

name: 'Glen 60cm Auto Clean Chimney',

brand: 'Glen',

type: 'Kitchen Chimney',

price: 12999,

image: 'https://glenindia.com/cdn/shop/files/1_b19848cb-8512-4162-92ff-a6e6638542a3_1400x1400.jpg?v=1712896874',

features: [

'60cm width, 1200 m³/hr suction power',

'Auto clean function',

'Touch control panel',

```

    '2-year warranty'
  ]
},
{
  id: 11,
  name: 'Hafele 4 Burner Gas Stove',
  brand: 'Hafele',
  type: 'Gas Stove',
  price: 63000,
  image: 'https://decure.in/cdn/shop/files/hob_haf_altius_plus_matt_glass_490_355cef74-4f2d-427e-b891-fc6cdffaae1d.jpg?v=1735555604&width=823',
  features: [
    '4 brass burners with auto ignition',
    'Premium mat finish body',
    'European design',
    '2-year warranty'
  ]
},
{
  id: 12,
  name: 'Crompton 60cm Kitchen Chimney ',
  brand: 'Crompton',
  type: 'Kitchen Chimney',
  price: 26000,
  image: 'https://www.crompton.co.in/cdn/shop/files/sensosmart-curv-baff-60-png.png?v=1728473934&width=1100',
  features: [
    '60cm width, 1416 m³/hr suction power',
    'Intelligent Heat Auto Clean with 30 hr timer',
    'Noise Level 50 dB',

```

```

    '1-year warranty'
  ]
}
];

// Initialize cart and user data
let cart = [];
let isLoginMode = true;
let currentUser = null;
let ordersPlaced = 0;
let activeBrandFilter = 'all';

// Function to render products
function renderProducts() {
  const productList = document.getElementById('product-list');
  productList.innerHTML = "";

  let filteredProducts = products;
  if (activeBrandFilter !== 'all') {
    filteredProducts = products.filter(product => product.brand.toLowerCase() ===
activeBrandFilter);
  }

  filteredProducts.forEach(product => {
    const productItem = document.createElement('div');
    productItem.className = 'product-item';

    let featuresHTML = "";
    if (product.features && product.features.length > 0) {
      featuresHTML = '<div class="features"><ul>';
      product.features.forEach(feature => {

```

```

    featuresHTML += `<li>${feature}</li>`;
  });
  featuresHTML += '</ul></div>';
}

```

```

productItem.innerHTML = `
  
  <div class="brand">${product.brand}</div>
  <h3>${product.name}</h3>
  ${featuresHTML}
  <p>₹${product.price.toLocaleString('en-IN')}</p>
  <button onclick="addToCart(${product.id})">Add to Cart</button>
`;
productList.appendChild(productItem);
});
}

```

// Function to render products on the home page

```

function renderHomeProducts() {
  const homeProductList = document.getElementById('home-product-list');
  homeProductList.innerHTML = "";

```

// Show only 6 featured products on home page

```

const featuredProducts = products.slice(0, 6);

```

```

featuredProducts.forEach(product => {
  const productItem = document.createElement('div');
  productItem.className = 'product-item';

```

```

  let featuresHTML = "";

```

```

if (product.features && product.features.length > 0) {
  featuresHTML = '<div class="features"><ul>';
  // Show only first 2 features on home page
  product.features.slice(0, 2).forEach(feature => {
    featuresHTML += `<li>${feature}</li>`;
  });
  featuresHTML += '</ul></div>';
}

```

```

productItem.innerHTML = `
  
  <div class="brand">${product.brand}</div>
  <h3>${product.name}</h3>
  ${featuresHTML}
  <p>₹${product.price.toLocaleString('en-IN')}</p>
  <button onclick="addToCart(${product.id})">Add to Cart</button>
`;
homeProductList.appendChild(productItem);
});
}

```

```

// Function to filter products by brand
function filterProducts(brand) {
  activeBrandFilter = brand;

  // Update active button styling
  document.querySelectorAll('.brand-filter button').forEach(btn => {
    btn.classList.remove('active');
  });
  event.target.classList.add('active');
}

```

```

    renderProducts();
  }

  // Function to add product to cart
  function addToCart(productId) {
    const product = products.find(p => p.id === productId);
    cart.push(product);
    updateCart();
    alert(`${product.name} added to cart!`);
  }

  // Function to remove product from cart
  function removeFromCart(productId) {
    const index = cart.findIndex(item => item.id === productId);
    if (index !== -1) {
      cart.splice(index, 1);
      updateCart();
    }
  }

  // Function to update cart display
  function updateCart() {
    const cartItems = document.getElementById('cart-items');
    const cartTotal = document.getElementById('cart-total');
    cartItems.innerHTML = "";

    let total = 0;
    cart.forEach(item => {
      const cartItem = document.createElement('div');

```

```

    cartItem.className = 'cart-item';
    cartItem.innerHTML = `
      <span class="item-name">${item.name}</span>
      <span class="item-price">₹${item.price.toLocaleString('en-IN')}</span>
      <button onclick="removeFromCart(${item.id})">Remove</button>
    `;
    cartItems.appendChild(cartItem);
    total += item.price;
  });

```

```

    cartTotal.textContent = `Total: ₹${total.toLocaleString('en-IN')}`;
  }

```

```

// Function for checkout

```

```

function checkout() {
  if (cart.length === 0) {
    alert('Your cart is empty!');
  } else {
    if (!currentUser) {
      alert('Please login to proceed with checkout.');
      showSection('auth');
    } else {
      const total = cart.reduce((sum, item) => sum + item.price, 0);
      const productList = cart.map(item => item.name).join(', ');

```

```

// Log entry for UI

```

```

    const logEntry = `Email: ${currentUser}\nProducts: ${productList}\nTotal:
    ₹${total.toLocaleString('en-IN')}\n-----\n`;

```

```

// Append to log display textarea

```

```

const logArea = document.getElementById('checkout-log');
logArea.value += logEntry;

// ✔ Send to backend (Node.js server)
fetch('http://localhost:3000/checkout', {
  method: 'POST',
  headers: { 'Content-Type': 'application/json' },
  body: JSON.stringify({
    email: currentUser,
    products: cart.map(item => item.name),
    total: total
  })
})
.then(res => res.text())
.then(msg => console.log('✔ Server response:', msg))
.catch(err => console.error('✖ Error sending to server:', err));

alert('Order placed successfully!\nTotal: ₹${total.toLocaleString('en-IN')}\nThank you
for your purchase.');
```

```

ordersPlaced++;
document.getElementById('order-count').textContent = ordersPlaced;
cart = [];
updateCart();
}
}
}

```

// Search functionality

```

function searchProduct() {
  const query = document.getElementById('search-input').value.toLowerCase();
  const filteredProducts = products.filter(product =>
    product.name.toLowerCase().includes(query) ||
    product.brand.toLowerCase().includes(query) ||
    product.type.toLowerCase().includes(query)
  );
  renderFilteredProducts(filteredProducts);
}

```

```

function renderFilteredProducts(filteredProducts) {
  const searchResults = document.getElementById('search-results');
  searchResults.innerHTML = "";

  if (filteredProducts.length === 0) {
    searchResults.innerHTML = '<p>No products found matching your search.</p>';
    return;
  }

```

```

  filteredProducts.forEach(product => {
    const productItem = document.createElement('div');
    productItem.className = 'product-item';

    let featuresHTML = "";
    if (product.features && product.features.length > 0) {
      featuresHTML = '<div class="features"><ul>';
      product.features.slice(0, 2).forEach(feature => {
        featuresHTML += '<li>${feature}</li>';
      });
      featuresHTML += '</ul></div>';
    }

```

```

    }

    productItem.innerHTML = `
        
        <div class="brand">${product.brand}</div>
        <h3>${product.name}</h3>
        ${featuresHTML}
        <p>₹${product.price.toLocaleString('en-IN')}</p>
        <button onclick="addToCart(${product.id})">Add to Cart</button>
    `;

    searchResults.appendChild(productItem);
});
}

// Function to switch between sections
function showSection(sectionId) {
    document.querySelectorAll('.section').forEach(section =>
    section.classList.remove('show'));

    document.getElementById(sectionId).classList.add('show');
    hideProfile();

    // Reset search results when showing home section
    if (sectionId === 'home') {
        document.getElementById('search-results').innerHTML = "";
        document.getElementById('search-input').value = "";
    }
}

// Function to show profile
function showProfile() {

```

```
const profileSection = document.getElementById('profile-section');
profileSection.style.display = profileSection.style.display === 'block' ? 'none' : 'block';
}
```

```
// Unified login/sign-in function
```

```
function handleAuth(event) {
```

```
  event.preventDefault();
```

```
  const email = document.getElementById('auth-email').value;
```

```
  const password = document.getElementById('auth-password').value;
```

```
  if (isLoginMode) {
```

```
    // Simple login validation
```

```
    if (email && password) {
```

```
      currentUser = email;
```

```
      document.getElementById('profile-email').textContent = email;
```

```
      document.getElementById('account-type').textContent = 'Registered Customer';
```

```
      alert(`Welcome back! You are now logged in as ${email}`);
```

```
      showSection('home');
```

```
    } else {
```

```
      alert('Please enter both email and password.');
```

```
    }
```

```
  } else {
```

```
    // Registration
```

```
    if (email && password) {
```

```
      currentUser = email;
```

```
      document.getElementById('profile-email').textContent = email;
```

```
      document.getElementById('account-type').textContent = 'Registered Customer';
```

```
      alert(`Thank you for registering! You are now logged in as ${email}`);
```

```
      showSection('home');
```

```
    } else {
```

```

        alert('Please complete all registration fields.');
```

```

    }
}

document.getElementById('auth-form').reset();
}

// Toggle between login and sign-in
function toggleAuth() {
    isLoginMode = !isLoginMode;
    const title = document.getElementById('auth-title');
    const button = document.getElementById('auth-button');
    const toggleLink = document.querySelector('.toggle-link');

    if (isLoginMode) {
        title.textContent = 'Login';
        button.textContent = 'Login';
        toggleLink.textContent = "Don't have an account? Register";
    } else {
        title.textContent = 'Register';
        button.textContent = 'Register';
        toggleLink.textContent = 'Already have an account? Login';
    }

    document.getElementById('auth-form').reset();
}

// Logout function
function logout() {
    currentUser = null;

```

```
document.getElementById('profile-email').textContent = 'guest@example.com';
document.getElementById('account-type').textContent = 'Guest';
alert('You have been logged out.');
```

```
showSection('auth');
hideProfile();
}
```

// Hide profile section

```
function hideProfile() {
  const profileSection = document.getElementById('profile-section');
  profileSection.style.display = 'none';
}
```

// Handle selling a product

```
function handleSell(event) {
  event.preventDefault();
  const name = document.getElementById('sell-name').value;
  const brand = document.getElementById('sell-brand').value;
  const type = document.getElementById('sell-type').value;
  const price = parseFloat(document.getElementById('sell-price').value);
  const image = document.getElementById('sell-image').value;
  const desc = document.getElementById('sell-desc').value;
```

// Extract features from description (simple approach)

```
const features = desc.split('\n').filter(line => line.trim() !== "");
```

```
const newProduct = {
  id: products.length + 1,
  name: name,
  brand: brand,
```

```
type: type,  
price: price,  
image: image,  
features: features  
};
```

```
products.push(newProduct);  
renderProducts();  
renderHomeProducts();  
alert(`Product "${name}" added successfully!`);  
document.getElementById('sell-form').reset();  
}
```

```
// Initialize the page  
renderProducts();  
renderHomeProducts();  
showSection('home');
```

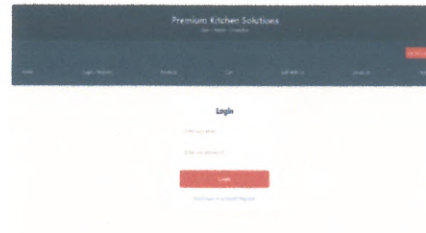
5. Results

5.1 Output Screens

Test case-1



Test case-2



Test case-3



Test case-4



Test case-5



6. Conclusion

To create a simplified e-commerce website using HTML, CSS, and JavaScript, begin by developing a basic site where users can browse products and add them to a shopping cart. Structure the website's content with HTML, apply a clean and responsive design using CSS, and implement interactive features like adding products to the cart with JavaScript. Key features should include displaying products with images, names, and prices, as well as allowing users to add products, view the cart, and modify quantities. For a practical guide, which offers a step-by-step walkthrough along with source code.

6.1 References

- **HTML:** Used for structuring the web page.
- **CSS:** Used for styling and responsive design.
- **JavaScript:** Used for dynamic interaction, such as searching through the movie list and filtering results based on user input.
- **Information:** Glen, crompton, Hafele (Websites)